NUCLEAR CHEMISTRY 2

24

Detecting radiation There are many ways to detect ionising radiation. Two of these ways are with photographic film and with the Geiger–Müller tube.

Workers who could be exposed to radiation often wear **dosimeters** (dosemeters) which are usually a photographic film badge of the type shown in figure 1. This can show how much radiation the wearer has been exposed to over a given period of time (or *doses* of radiation, hence the name). The example shown has a piece of photographic film wrapped in a light-proof cover and placed inside a container that can be clipped to worker's clothing. The film is developed at regular intervals.

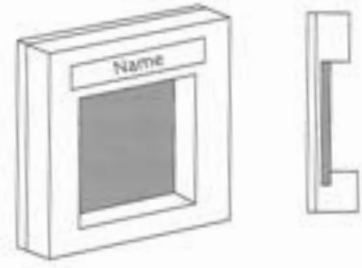


Figure 1

1 Copy figure 1 and label the following: *plastic container, light-proof covering, photographic film.*

2 Explain how the developed film in the dosimeter indicates the level of radiation exposure of a worker.

Film badges show the cumulative radiation a worker has been exposed to over a period of time, but in many cases we need to know how much radiation is currently being emitted. A **Geiger–Müller tube and counter** (usually referred to as a **Geiger counter**) tells how much radiation is given off by a source in 'real time'.

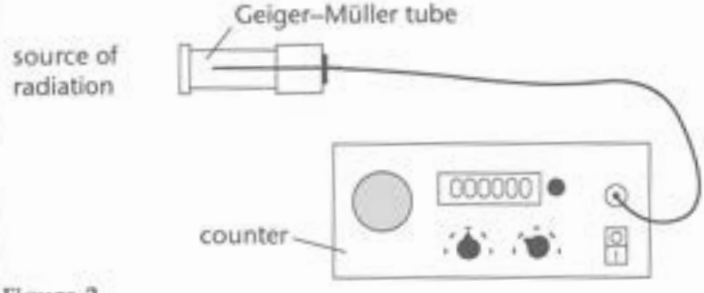


Figure 2

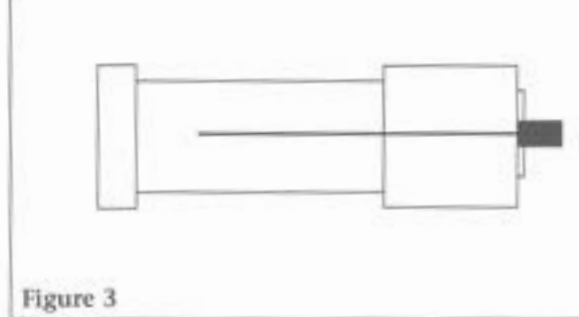

Figure 3

Figure 2 shows how a Geiger counter is set up to detect and measure radiation. Figure 3 is a simplified view of a Geiger–Müller tube. The tube consists of an outer casing with a 'window' at the front of the tube that allows radiation to pass through. The inside has a central wire at a potential of around + 400 volts surrounded by argon gas. When ionising radiation enters the tube, it strips an electron from an argon atom. The electron (having a negative charge) is attracted to the central wire. As it accelerates towards the positive charge, the electron collides with other argon atoms which causes more electrons to be released. The group of electrons arrives at the wire which causes a pulse of current that is detected, amplified and counted.

3 Copy figure 3. Use the information from the text to label the parts of the tube.

4 Outline the advantages and disadvantages of detecting radiation with a Geiger counter compared to a photographic film badge.

Sources of radioactivity

5 Some sources of radioactivity are natural and others are made by humans. Mark each source of radiation as natural or caused by humans.

a Radiation from outer space, e.g. cosmic rays

b Waste from nuclear power stations

c Radioisotopes deep in the Earth's core

d 'Fallout' from nuclear explosions

Half-life A source of radioactivity cannot give out energy and particles forever. As time goes by the amount emitted must decline. The concept of a 'half-life' is used to describe how long a substance will remain radioactive.

The **half-life** of a radioisotope is the average time it takes for half of the radioactive atoms to decay to a different atom.

6 Use a textbook to give another definition of *half-life*.

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The half-life of a radioisotope varies from a fraction of a second to over a million years, depending on the particular radioisotope. Knowing the half-life of a radioisotope can help forecast how long it will remain a cause of concern. It can also help determine the age of artefacts. Iodine–131 has a half-life of 8 days. This means that after 8 days the radiation will be half of the original radioisotope; after 16 days, it will be a quarter; after 24 days, an eighth, and so on.

7 What percentage of the original iodine–131 isotope would be present after 40 days?

Carbon–14 is a naturally-occurring radioisotope present at constant levels in all living creatures. This radioisotope has a half-life of around 5 800 years. When the organism dies, the level of carbon–14 declines. By comparing the amount of carbon–14 present in an archaeological specimen with that in a living example, the age of the specimen can be calculated. This is called radiocarbon dating.

8 How much of the original carbon–14 will be in the remains of a tree after 11 600 years?

9 A skeleton has one-sixteenth of the carbon–14 of a living equivalent. How old is it likely to be?

10 A piece of pottery is thought to be around 10 000 years old. Would radiocarbon dating be useful in establishing how old this object is? Explain your answer.

Uses of radioisotopes

The use of a radioactive isotope depends on the penetrating power of its radiation and the length of its half-life. Some applications need a steady source of radiation (long half-life) and others need radiation that diminishes in strength quickly.

Alpha particle sources Alpha particles only penetrate weakly. Smoke detectors tend to use a source of alpha radiation that sends particles to a detector across a gap of air. When smoke particles are present, this reduces the number of alpha particles reaching the detector and electronic circuitry can generate an alarm.

Beta radiation sources Beta particles are usually stopped by a solid material a few centimetres thick. The amount of radiation absorbed by the material is proportional to the thickness (and density) of the material. The thicker the material, the more beta radiation it absorbs. Materials that are produced in sheet form to absorb certain amounts of radiation can have their thicknesses monitored. The sheet passes between a beta source and a detector. If the amount of radiation detected is too high, the sheet is too thin; if the radiation is too low, the sheet is too thick. This information is fed back to the production machinery to alter thickness.

Gamma radiation sources Gamma radiation is the most highly penetrating and is used when radiation must be able to pass through a substantial thickness of a material. It can be used as a tracer; e.g. to detect leaks in pipes carrying fluids and, after injection, to follow biological processes in the human body. For the latter use, the half-life should be as short as possible to minimise any health hazards. The destructive nature of gamma radiation can be used to kill cancerous cells (radiotherapy) and to sterilise food and surgical instruments. The gamma emitters in the instruments used to destroy cells and bacteria need to have relatively long half-lives to allow the instruments to work for a reasonable length of time.

11 Draw up a table to summarise the uses of radiation. Use the headings below.

Type of radiation	Function	Half-life	Penetrating power

12 Give another way radiotherapy is used and briefly describe how it works.

13 Describe how technicians who work in radiotherapy clinics might be put at risk and how these risks can be minimised.

14 Match each term to the correct defintion: *radioisotope, gamma radiation, beta radiation, isotope, alpha radiation.*

a High-speed electrons given off during radioactive decay

b High-energy (short-wavelength) radiation emitted during radioactive decay

c A radioactive isotope

d The nucleus of a helium atom, given off during radioactive decay

e An element with the same number of protons but a different number of neutrons

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